

isBlacklisted

Takes the client id as input, checks if the client is in the blacklist table, and returns true or false.

```
CREATE OR REPLACE FUNCTION isBlackListed(c_id CLIENT.C_ID%type) RETURN
BOOLEAN
IS
    clients NUMBER;
BEGIN
    SELECT COUNT(*) INTO clients FROM BLACK_LIST WHERE c_id = C_ID;
    RETURN clients > 0;
END;
/
```

```
Function ISBLACKLISTED compiled
Client is blacklisted.
```

isIssued

Takes vehicle id, start and end date, and returns if the vehicle has been issued within that time frame.

```
CREATE OR REPLACE FUNCTION isIssued(v_id IN VEHICLE.V_ID%TYPE, start_date
IN DATE, end_date IN DATE) RETURN BOOLEAN
IS
    conflicts NUMBER;
BEGIN
    SELECT COUNT(*) INTO conflicts
    FROM RESERVATION
    WHERE V_ID = v_id
        AND (
            (start_date BETWEEN START_DATE AND END_DATE) OR
            (end_date BETWEEN START_DATE AND END_DATE) OR
            (START_DATE BETWEEN start_date AND end_date) OR
            (END_DATE BETWEEN start_date AND end_date)
        );
    RETURN conflicts > 0;
END;
/
```

```
Function ISSUED compiled
Vehicle is issued.
```

calculateTotalRevenue

Takes the vehicle id and sums up the revenue from each reservation on that vehicle. This satisfies question 5.

```
CREATE OR REPLACE FUNCTION calculateTotalRevenue(v_id VEHICLE.V_ID%type)
RETURN NUMBER
IS
  v_sum_res NUMBER;
  v_sum_arch NUMBER;
BEGIN
  SELECT SUM(TOTAL_COST) INTO v_sum_res FROM RESERVATION WHERE v_id =
  V_ID;
  SELECT SUM(TOTAL_COST) INTO v_sum_arch FROM ARCHIVED_RESERVATION
  WHERE v_id = V_ID;
  RETURN v_sum_res + v_sum_arch;
END;
/
```

Function CALCULATETOTALREVENUE compiled

Vehicle revenue: 3480

returnCar & calculateTotal

Function for returning a car, return car takes a reservation id, client id, is paid and amount returned. Return car calls calculate total. Return car prints out the total cost owed after payment.

```
CREATE OR REPLACE FUNCTION calculateTotal (
    start_date DATE,
    end_date DATE,
    v_rate VEHICLE.COST_PER_DAY%TYPE
) RETURN NUMBER
IS
    christmas DATE := TO_DATE('25-DEC', 'DD-MON');
    rate NUMBER := 1;
    total_days NUMBER;
BEGIN
    total_days := end_date - start_date + 1;
    IF total_days > 10 THEN
        rate := rate - 0.10;
    END IF;

    IF christmas BETWEEN start_date AND end_date THEN
        rate := rate - 0.20;
    END IF;
    RETURN rate*v_rate*total_days;
END;
/

CREATE OR REPLACE PROCEDURE returnCar(
    reservation_id IN RESERVATION.R_ID%TYPE,
    client_id IN CLIENT.C_ID%TYPE,
    is_paid IN NUMBER,
    car_damaged IN NUMBER
) AS
    start_date DATE;
    end_date DATE;
    client_id_res CLIENT.C_ID%TYPE;
    daily_rate VEHICLE.COST_PER_DAY%TYPE;
    total_cost NUMBER;
BEGIN
    IF reservation_id IS NOT NULL THEN
        SELECT START_DATE, END_DATE, C_ID, TOTAL_COST INTO start_date,
end_date, client_id_res, daily_rate
        FROM RESERVATION
```

```

        WHERE R_ID = reservation_id;
    ELSE
        SELECT START_DATE, END_DATE, C_ID, TOTAL_COST INTO start_date,
end_date, client_id_res, daily_rate
        FROM RESERVATION
        WHERE C_ID = client_id
        ORDER BY END_DATE DESC
        FETCH FIRST 1 ROWS ONLY;
    END IF;

    total_cost := calculateTotal(start_date, end_date, daily_rate);

    IF is_paid = 0 OR car_damaged = 1 THEN
        INSERT INTO BLACK_LIST (C_ID, F_NAME, L_NAME, DOB, CITY, GENDER,
REASON, ADDED_BY, ADDED_DATE)
        SELECT C_ID, F_NAME, L_NAME, DOB, CITY, GENDER,
            CASE
                WHEN is_paid = 0 THEN 'Non-payment'
                WHEN car_damaged = 1 THEN 'Damaged vehicle'
            END,
            'system', SYSDATE
        FROM CLIENT
        WHERE C_ID = client_id_res;

        DBMS_OUTPUT.PUT_LINE('Client has been added to the blacklist for: '
||
                                CASE
                                    WHEN is_paid = 0 THEN 'Non-payment'
                                    WHEN car_damaged = 1 THEN 'Damaged
vehicle'
                                END);
    ELSE
        DBMS_OUTPUT.PUT_LINE('Total cost after discount: $' || total_cost);
    END IF;
END;
/

```

Procedure RETURN_CAR compiled

Total cost after discount: \$8400